Polynomial Techniques

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1 Polynomial Division

Polynomials are the templates of algebra. They allow us to capture the major operations of adding, subtracting, and multiplying all in the same expression. For example, an expression like $x^2 - x + 2$ makes good sense for any number we want to put in for x. Division is generally not included in expressions like these. For example the expression

$$x^2 - \frac{1}{x} + 2,$$

while algebraic in nature, is not considered a *polynomial*. One reason why division is handled on its own, is because division does *not* work for all numbers x. The above expression for example, does not make sense at $x = 0$ (since we cannot divide by zero).

Though we could run into troubles when dividing by inadmissible values for x, it can still make good sense to *divide polynomials*. Quotients of polynomials are natural objects, as they encompass all four algebraic/arithmetic operations (at the cost of omitting some values of x where it makes sense). But aside from their inherent interest, they can be incredibly informative to tell us how polynomials *behave*.

In what follows, we talk about a couple of strategies for dividing polynomials, such as *long division*. We finish off by catching a glimpse at what these divisions tell us about our polynomials.

1.1 Long Division

If multiplication is a repeated addition,

$$2 \times 3 = 2 + 2 + 2 \quad (\text{or} \;\; 3 + 3),$$

then division is a repeated *subtraction* in the following sense. Say we want to divide 21 by 3. We repeatedly subtract 3 from 21 until we hit zero (or a sufficiently small number, more on this shortly):

$$21 - 3 - 3 - 3 - 3 - 3 - 3 - 3 = 0.$$

The number of times we have subtracted 3 from 21 to reach zero is our quotient. For example, in the above, it took us seven subtractions of 3 from 21 to get to zero, so $\frac{21}{3} = 7$.

There are some shortcuts that we naturally take when faced with more and more tedious calculations. Suppose for example we must divide 784 by 7. Rather than constantly subtracting by 7 over and over again

$$784 - 7 - 7 - \cdots - 7,$$

we might take out a whole chunk of 7's at once. For example, 784 has at least a hundred 7s that we can take from it, so why not just subtract these all at once, we are left with

$$784 - 100 \times 7 = 84.$$

Now we could proceed from here to subtract 7 one by one, or we could notice that there are at least another ten 7s that can be subtracted in one go. Subtracting these yield:

$$84 - 10 \times 7 = 14.$$

From here it is clear that we only need to subtract twice more to get to zero. So what is our quotient? We subtracted 100 + 10 + 2 sevens, for a total of 112. This tells us that $\frac{784}{7} = 112$.

The same principle guides us for dividing two polynomials. For example, suppose we want to find the quotient $(x^2-4)/(x-2)$. If we approach this in our original way of subtracting $x-2$ from x^2-4 one by one, then it looks like we will just make our problem worse:

$$(x^2 - 4) - (x - 2) = x^2 - x - 2,$$

subtracting again yields,

$$(x^2 - x - 2) - (x - 2) = x^2 - 2x,$$

we've gotten rid of our constant term, but this isn't exactly *simpler*... let's try one more

$$(x^2 - 2x) - (x - 2) = x^2 - 3x + 2.$$

Okay now we have our constant term back and we are just going to get more and more complicated from here. So how could we do this?

The trick is to take out *chunks* of $(x-2)$'s at once, similar to how we handled the large numbers in our numerators above, like 784/7. So what *chunk* of $(x-2)$'s can be removed from (x^2-4) at once to make it *simpler*? That is, what should we put in the box below to get a simpler polynomial remaining?

$$(x^2 - 4) - \boxed{} \times (x - 2)$$

First, what might *simpler* mean in this context? By *simpler* we will mean that our resulting polynomial has smaller *degree* (highest power of x in the polynomial). So is there a chunk of $(x-2)$'s that we can subtract from x^2-4 to remove the x^2? Hint: x is a placeholder for a number!

The trick then is to remove a chunk of $(x-2)$'s of size x! Let's see what happens if we do this:

$$(x^2 - 4) - x \times (x - 2) = (x^2 - 4) - (x^2 - 2x) = 2x - 4$$

Now we have successfully made our polynomial simpler, and if we continue with our usual subtraction method, we see that we only need to subtract $(x-2)$ twice more to get to zero:

$$(2x-4)-(x-2)-(x-2)=0.$$

So what is our quotient? Well we subtracted $(x-2)$ from x^2-4 a total of $x+2$ times, so this leaves us with $(x^2-4)/(x-2)=x+2$!

There are various tricks (such as the difference of squares) that could allow you to see the answer to the above quotient much quicker, however the procedure we just performed works in general and is called *polynomial long division*.

Let's do one more example before adding a complication. Suppose we want to find the quotient $(x^3-3x^2-9x+27)/(x-3)$. This one looks more daunting, but we can apply the same ideas of trying to reduce the degree by taking out chunks of $(x-3)$'s. The largest degree term is x^3, so we look for a chunk of $(x-3)$'s to cancel this term off.

$$(x^3-3x^2-9x+27)-\boxed{}\times(x-3)$$

It seems like a chunk of size x^2 could do the trick here.

$$\begin{aligned}(x^3-3x^2-9x+27)-x^2\times(x-3)&=(x^3-3x^2-9x+27)-(x^3-3x^2)\\&=-9x+27\end{aligned}$$

This got us down from a degree 3 polynomial to a degree 1 polynomial in one step! Then how many more $(x-3)$'s must we remove to get to zero? Here it looks like -9 more of them should do the trick:

$$(-9x+27)-(-9)\times(x-3)=0$$

So we removed a total of x^2-9 copies of $x-3$ from $x^3-3x^2-9x+27$ to get to zero, telling us our quotient is

$$(x^3-3x^2-9x+27)/(x-3)=x^2-9.$$

These examples have had the luxury that when we successively subtract, we always get to zero. However, most situations are not so simple, so we need to slightly amend the criterion we use to know that we have finished the process appropriately. For now though, try your hand at the problems below to get a feel for the long division process.

1.2 Exercises

For each of the following, find the quotient using long division (successively removing chunks of the denominator to reduce the polynomial to zero).

1. $(x^2-2x+1)/(x-1)$
2. $(x^2-4x+3)/(x-3)$
3. $(x^3+x^2-2x)/(x+2)$

We can even divide by degree 2 (or more) polynomials, it works the same way!

4. $(x^3 + 3x^2 + 3x + 2)/(x^2 + x + 1)$

Though the analogy of repeated subtraction works well for polynomials with whole number (integer) coefficients, it breaks down a bit when we work with fractional (rational) ones. However, the process can be directly adapted to this setting. Try using long division on the following (noting that we are aiming to reduce the degree of the polynomial at each stage).

5. $(\frac{1}{2}x^2 + x + \frac{1}{2})/(x + 1)$

As a simple curiosity, consider the following.

6. We have not been dividing simply by x. Why? Perform the long division $(x^3 + x^2 + x)/x$. Why was this one much easier? Can you find an analogy with dividing numbers that end in 0 by 10? (E.g. divide 23500/10.)

Try the following for a more theoretical problem:

7. Let a and b be two real numbers. Show (via long division) that

$$(x^2 - (a + b)x + ab)/(x - a) = (x - b)$$

and

$$(x^3 - (2a + b)x^2 + (2ab + a^2)x - a^2b)/(x - a) = (x^2 - (a + b)x + ab)$$

1.3 Remainders

It is not always the case that division leads to a nice round answer. For example, if we wanted to divide 14 by 5, we might repeatedly subtract,

$$14 - 5 - 5 = 4,$$

which leaves us in a bit of a strange spot. On one hand, there are still positive things left to divide, on the other if we subtract any more we will be left with negative numbers (and then we could subtract infinitely more from here). So in this case, we declare that we've subtracted enough (there's no more 5's to subtract), and we will say that we have a leftover or *remainder* of 4. Then we can write

$$14/5 = 2 + 4/5.$$

We will do something similar with polynomials that don't divide nicely. For example, consider the quotient $(x^2 + x + 1)/(x - 1)$. We perform long division the same way at the start; first subtract a chunk of $(x - 1)$'s to get rid of the x^2 term

$$(x^2 + x + 1) - x \times (x - 1) = x^2 + x + 1 - (x^2 - x) = 2x - 1$$

Next, we get rid of the $2x$ term,

$$(2x-1) - 2 \times (x-1) = 2x - 1 - (2x-2) = 1.$$

Here we are left in a similarly awkward spot, for if we try to now "remove" the remaining 1 leftover, we find that there is no reasonable chunk of $(x-1)$'s that can accomplish this feat, outside of using quotients. For example, maybe we try a chunk of size $\frac{1}{x}$:

$$1 - \frac{1}{x} \times (x-1) = 1 - (1 - \frac{1}{x}) = \frac{1}{x}.$$

But then next, we would need a chunk of size $\frac{1}{x^2}$ to cancel the leading term off here, leading us to an infinite task.

Instead, we find a natural stopping point when we get to the stage where we are left with just 1, and declare that since the degree of our leftover polynomial is smaller than that of our denominator, we have finished our long division, in this case with remainder 1. We can then write

$$(x^2 + x + 1)/(x-1) = (x+2) + \frac{1}{x-1}$$

Note the $x+2$ is coming from the fact that we subtracted a chunk of x $(x-1)$'s off first, and then two more.

Let's see another example. Suppose we want to divide $x^3 - 2x + 3$ by $x^2 - 4$. Proceeding with long division, we try to cancel off the leading x^3 by subtracting a chunk of x (x^2-4)'s:

$$(x^3 - 2x + 3) - x \times (x^2 - 4) = x^3 - 2x + 3 - (x^3 - 4x) = 2x + 3.$$

After just one step, we are left with a polynomial whose degree is smaller than that of our divisor, meaning it is time to stop with a remainder of $2x+3$. We can then write

$$(x^3 - 2x + 3)/(x^2 - 4) = x + \frac{2x+3}{x^2-4}.$$

Try some for yourself!

1.4 Exercises

Use polynomial long division to find the quotient and remainder of the following:

1. $(x^2+1)/(x-1)$

2. $(x^3+x^2+x+1)/(x-1)$

3. $(x^4-3x^2+1)/(x^2+1)$

2 Polynomial Equations

As soon as we are aware of how to add, subtract, and multiply numbers, evaluating polynomials is simple. For example, consider the polynomial $x^2 + x + 1$.

It is simple to evaluate this polynomial at any given value of x, for example at $x = 1$, we just perform the arithmetic $1^2 + 1 + 1 = 3$. At $x = 2$, we get $2^2 + 2 + 1 = 7$, and so on. Starting with an input, leads quickly and easily to an output.

However, suppose we observe an output of the polynomial. Is it possible to find what inputs lead to that output? This is the problem of solving *polynomial equations*. For example, suppose we wanted to find the input number x such that, when plugged into the polynomial $x^2 - 1$ yields the output 3. We can set this up as a simple equation:

$$x^2 - 1 = 3$$

and proceed to solve. In this case, it is simple enough to isolate x:

$$x^2 = 4$$

But then we run into our first bit of difficulty: there are two possible values that make this equation true! Indeed, taking $x = 2$ or $x = -2$ both yield the same output (check for yourself that both of these values satisfy the original equation). So we immediately need to abandon the idea that there is a unique input for a given output.

More complicated still, it's not so obvious that we can even solve for x. For example, if we want to find an input x such that

$$x^3 - 3x + 3x = 1$$

how might one even begin here? Try this for yourself! You might find it surprisingly difficult to isolate x and solve it in the same way as above. If you're curious, there is a unique real number x satisfying the above equation, can you find which one? (It turns out that there are also two additional complex valued solutions.)

This problem is clearly much more complicated than simply performing arithmetic. Luckily a bit of algebra knowledge can go a long way to helping us here.